

Deep Learning-Based Segmentation of Acute Stroke Lesions from Diffusion-Weighted MRI

Pawan Kumar Thapaliya
Quantitative MRI and Deep Learning intern
Moffitt Cancer Center, Tampa Florida
Advisor: Dr. Tess Armstrong
Jul-Oct 2025



- Motivation
- Background
- Methods
- Results
- Conclusion

Stroke Overview



The brain is 2% of body weight

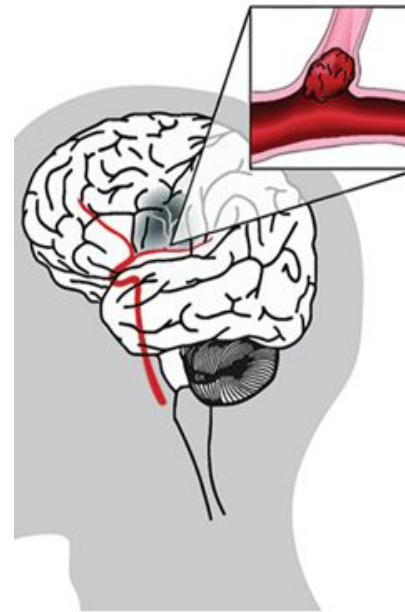
The brain consumes 20% of total energy

Stroke = interruption of blood flow to the brain → ↓ oxygen & ↓ nutrients

Types of Stroke:

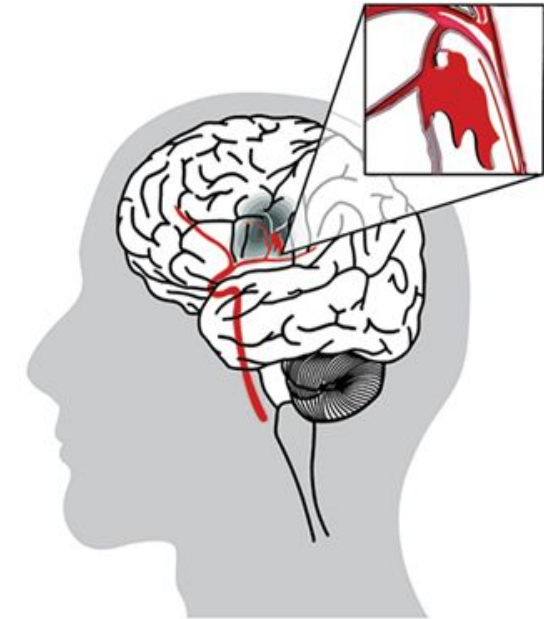
- **Ischemic:** vessel blockage (most common)
- **Hemorrhagic:** vessel rupture / bleeding

Block



Ischaemic stroke

Bleed

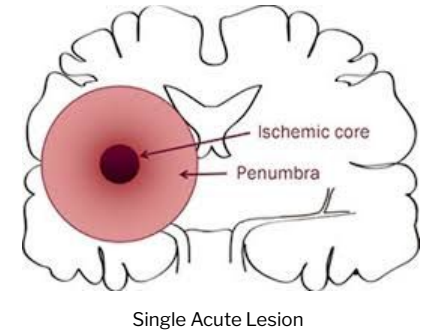


Haemorrhagic stroke

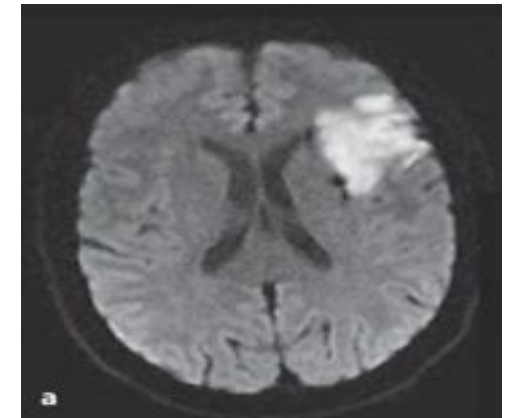
<https://enableme.org.au/resources/types-of-stroke>

Clinical Significance

- **2nd leading cause of death worldwide**
- **Major cause of long-term disability**
(Feigin et al., *Int J Stroke* 2022 — WSO Fact Sheet)



Single Acute Lesion



Schwarzbach et al., 2015

Problem: Requires **rapid diagnosis** → *time = brain*



<https://www.istockphoto.com/illustrations/hospital-death>

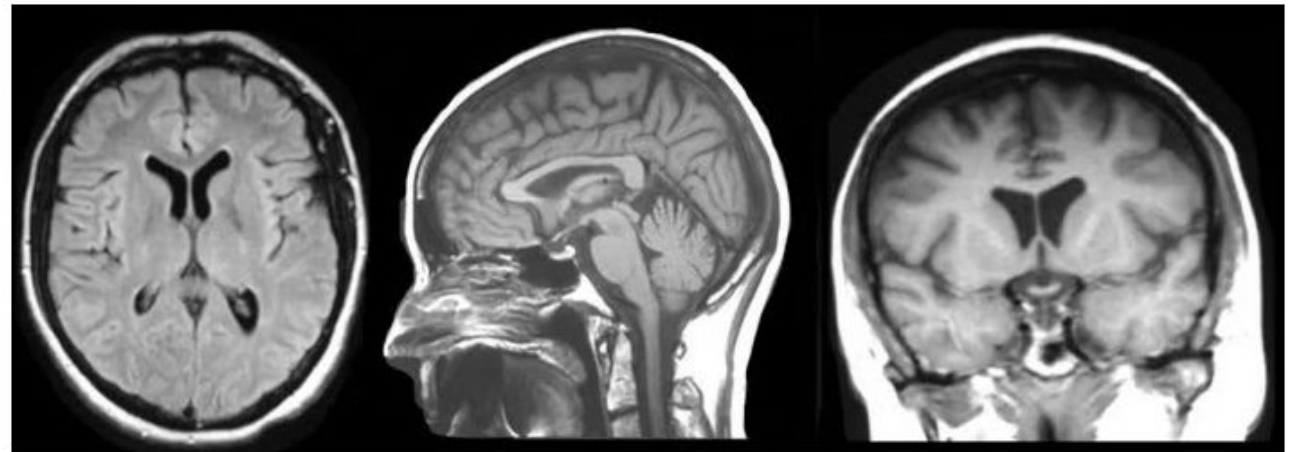
Role of Imaging



- Needed to differentiate stroke type
- Determines treatment eligibility + tissue salvage potential

e.g. CT, MRI, X-ray

Axial, Sagittal, and Coronal Views of Brain MRI



Source: <https://www.shutterstock.com>

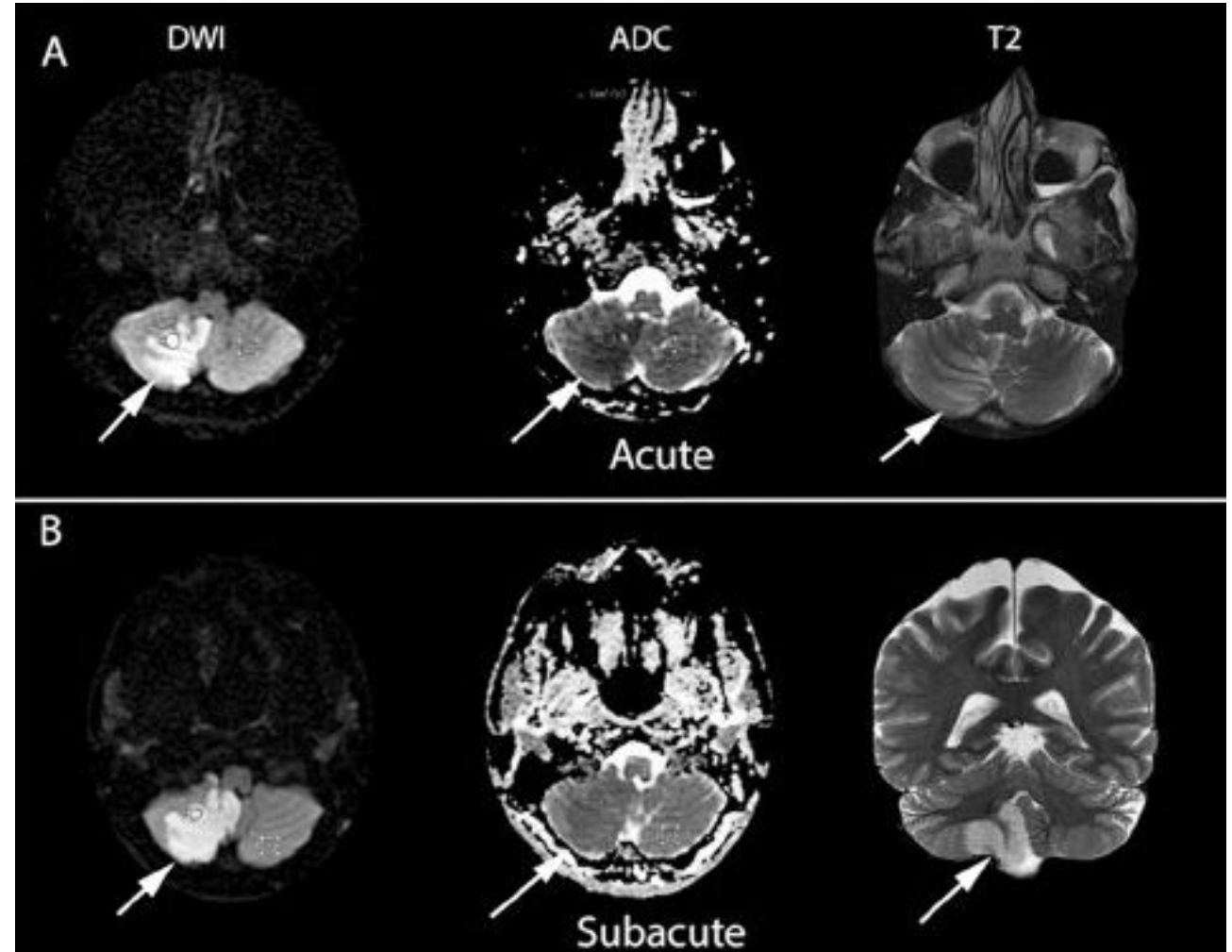
MRI

- **Non-invasive** and **non-ionizing** imaging method
- Provides **structural** and **functional** detail
- Can provide **quantitative** insight into human biology

Why MRI?



- **Diffusion (DWI)** → early ischemia detection (minutes)
- **Perfusion (PWI)** → evaluates blood flow & penumbra
- **T2 / FLAIR** → edema, age of lesion (tissue injury evolution)
- **T2* / SWI** → microbleeds + hemorrhage

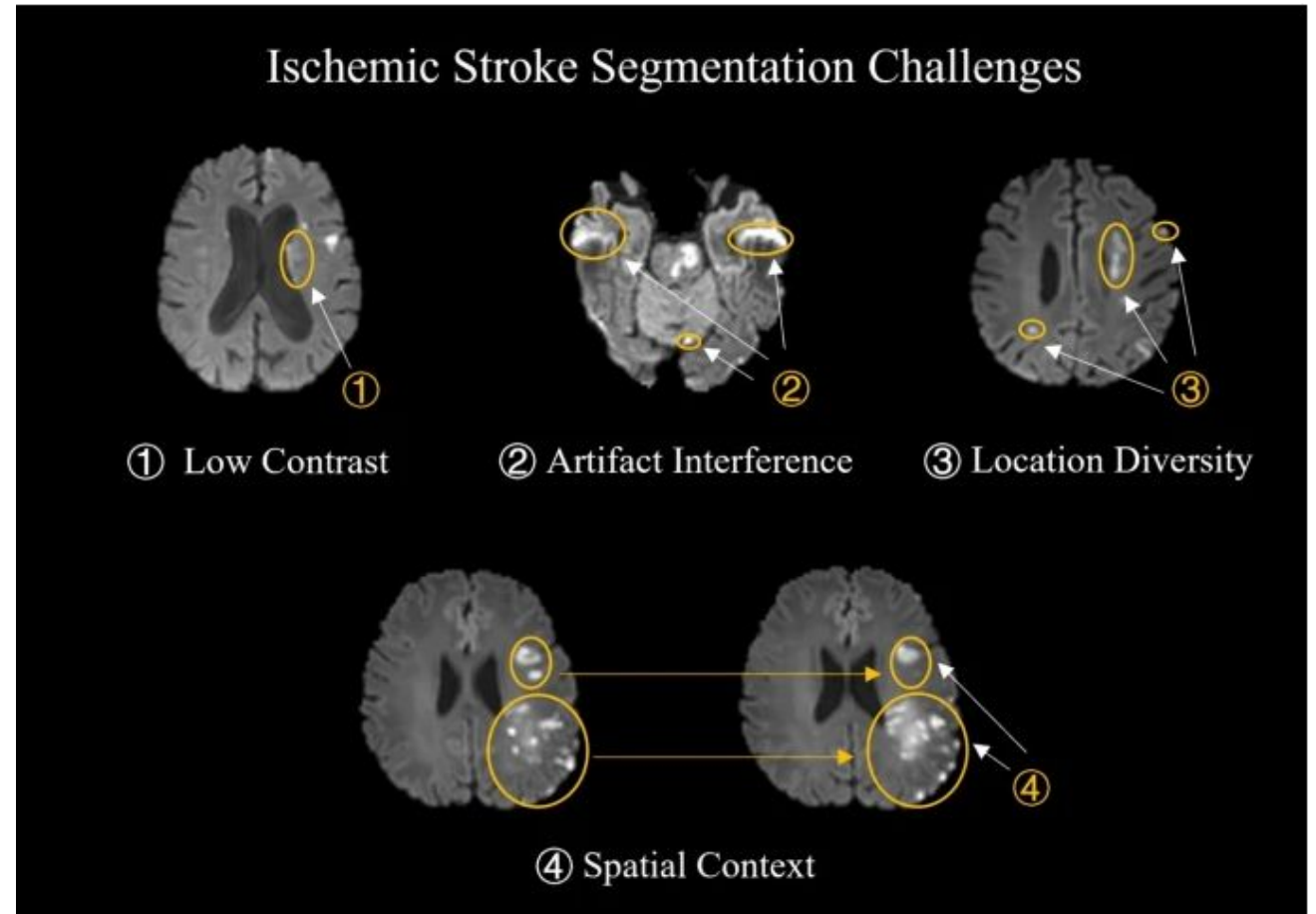


Roldan-Valadez et al., 2014

Challenges in Stroke Imaging

- Motion can cause image artifacts
- Variability in lesion size and location
- Partial volume effects

Problem: Need for automated or semi-automated segmentation



Chen et al., M2FNet (2023)

Why Automated Segmentation?



Manual interpretation is slow, variable, and requires expert neuroradiologists

- Enables **objective** and **reproducible** lesion identification
- Supports rapid assessment in acute stroke workflow
- Converts MRI into **quantitative maps** instead of qualitative visual inspection
- Improves treatment decision-making + outcome prediction
- Foundation step for ML / deep learning stroke analysis pipelines

Goal → transform MRI into measurable, consistent stroke biomarkers through segmentation



Classical vs. Deep Learning Segmentation

Classical:

- Thresholds / Atlas / Region Growing
- Hand-crafted features
- Limited cross-scanner generalization
- Requires manual tuning
- Slower + struggles with complex stroke tissue

Deep Learning:

- CNNs / U-Net / Transformer based models
- Learns features automatically
- More robust + domain adaptable
- End-to-end learning pipeline
- Fast inference + higher accuracy in stroke lesions

Deep Learning offers **scalable, reliable, clinically meaningful** segmentation.

Background: Ischemic Stroke MRI



- Ischemic stroke lesions are **subtle and variable**, making accurate MRI segmentation challenging.
- DWI, ADC, and eDWI** improve lesion detection; multi-modal integration boosts accuracy.
- Deep learning methods using **DenseNet121 + SelfONN + attention** outperform traditional approaches.
- ISLES 2022 benchmark:** DWI alone DSC 83.9%, DWI+ADC 85.9%, DWI+ADC+eDWI 87.5%.

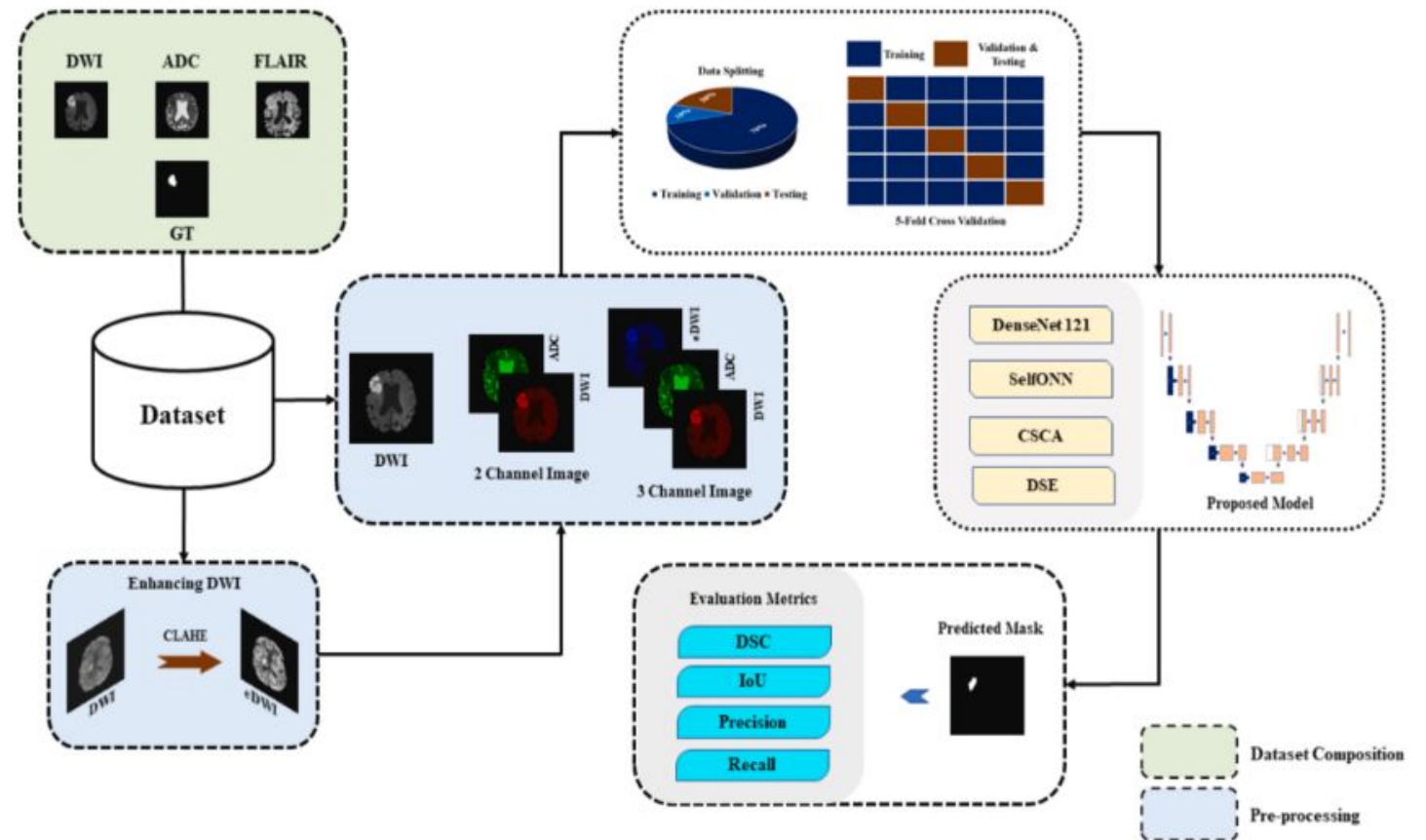
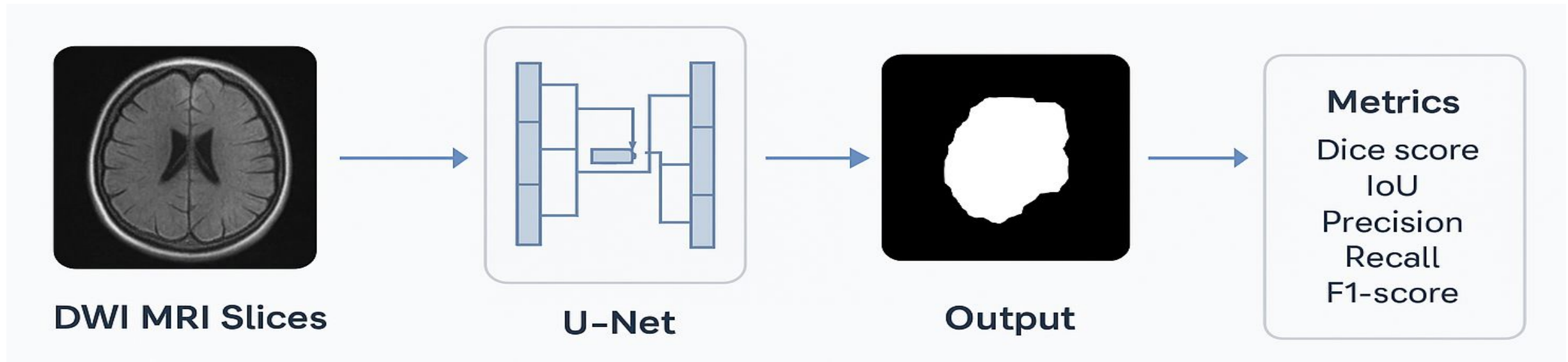


Fig. 1. Schematic representation of the experimental pipeline for ischemic lesion segmentation.

Rahman et al., 2025, Bio-Signal Process. Control – DSC up to 87.5%

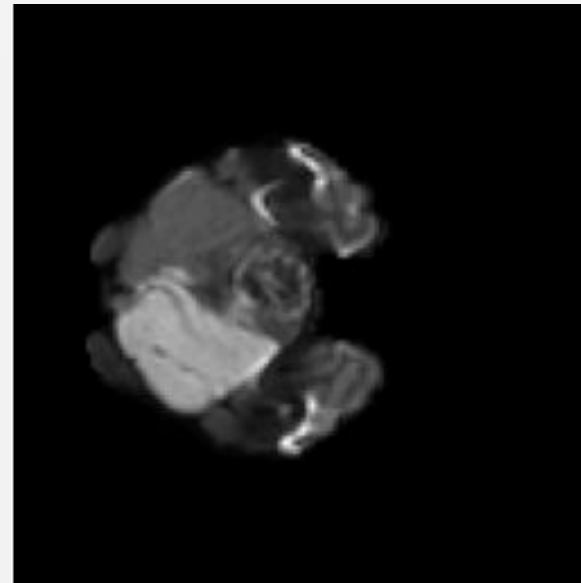
Methods



Data Set Overview

- Kaggle dataset:
<https://www.kaggle.com/datasets/muhhilmy/only-slices-with-stroke>
- **Modalities:** DWI MRI slices + Binary masks
- Grayscale(1 channel), resized to 128 X 128
- **Number of cases:** Train: 3378, Val: 724, Test: 725
- **Train/Val/Test split:** 70% / 15% / 15%

DWI Stroke Slice



Mask





Normalization

- Mean & standard deviation from training images
- Normalize images using $(x - \mu) / \sigma$
- Masks kept binary

Model Architecture (U-Net):

- Input: 1-channel grayscale image
- Kernel size: 3×3 (padding=1, stride=1)
- Encoder filters: 64 → 128 → 256 → 512 → 1024
- Decoder: Transposed conv + skip connections
- Group Normalization, ReLU activation
- Sigmoid final layer → 1-channel probability mask

Training Setup

- Optimizer: Adam
(learning rate = $1e-3$)
- Loss function: BCE + Dice loss
- Batch size: 8, Epochs: 50
- GPU acceleration when available

Evaluation Metrics

- Dice Score
- Intersection over Union (IoU)
- Precision
- Recall
- F1-score

Monitoring

- Train/Validation Loss curves
- Train/Validation Dice curves

Confusion Matrix, Precision, Recall, F1-score



	Predicted Positive	Predicted Negative
Actual Positive	TP	FN
Actual Negative	FP	TN

TP = true positives

FP = false positives

FN = false negatives

TN = true negatives

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

$$\text{F1-score} = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

$$\text{F1-score} = \frac{2TP}{2TP + FP + FN}$$

Dice Similarity Coefficient (DSC)



Set form:

$$\text{DSC} = \frac{2|P \cap G|}{|P| + |G|}$$

TP/FP/FN form:

$$\text{DSC} = \frac{2TP}{2TP + FP + FN}$$

Dice Loss:

$$\mathcal{L}_{\text{dice}} = 1 - \text{DSC}$$

P = prediction mask

G = ground truth mask

Dice Loss ensures the predicted mask overlaps well with the ground truth (shape/region-level)

Binary Cross Entropy Loss



Single sample:

$$L_i = -[y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)]$$

Batch of N samples:

$$L = -\frac{1}{N} \sum_{i=1}^N [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)]$$

Special cases:

$$y_i = 1 \Rightarrow L_i = -\log(\hat{y}_i)$$

$$y_i = 0 \Rightarrow L_i = -\log(1 - \hat{y}_i)$$

Symbols:

- ▶ N : number of samples
- ▶ x_i : input vector
- ▶ $y_i \in \{0, 1\}$: true label
- ▶ $z_i = w^\top x_i + b$: logit
- ▶ $\hat{y}_i = \sigma(z_i)$: predicted probability
- ▶ $\sigma(z) = \frac{1}{1 + e^{-z}}$: sigmoid
- ▶ w, b : learnable parameters
- ▶ η : learning rate

BCE ensures each pixel is predicted correctly (pixel-wise accuracy)

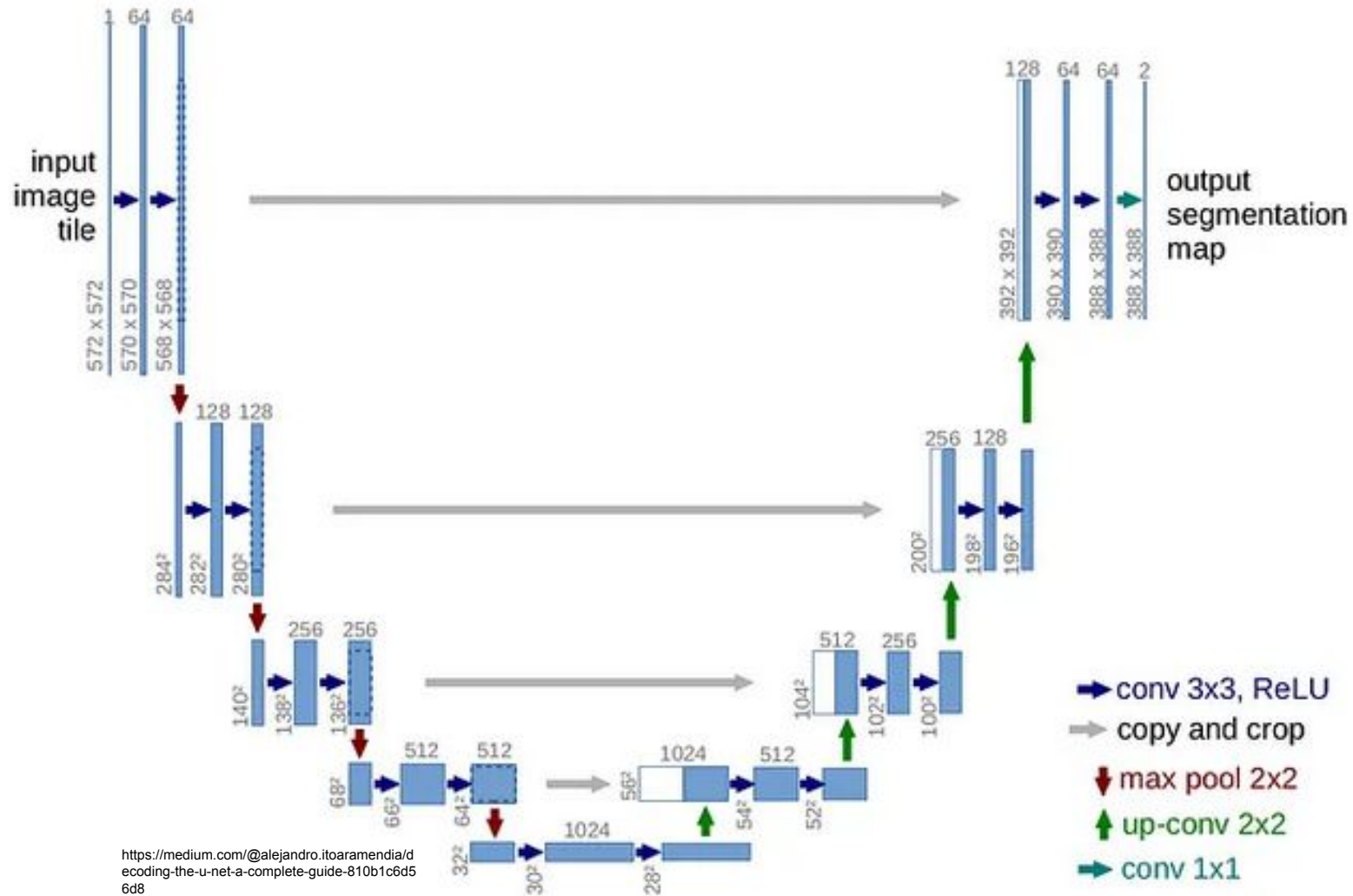


Total Loss = BCE + Dice Loss

Combined loss balances accuracy and overlap for better segmentation

Forward Pass $\Rightarrow$ Loss $\Rightarrow$ Backprop $\Rightarrow$ update

Unet Architecture



Results of Standard Unet Model

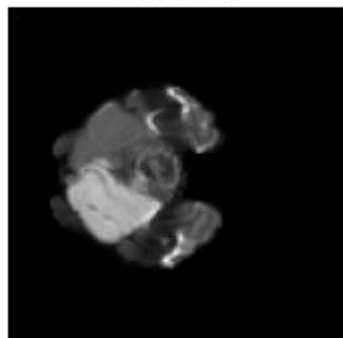


=== Test metrics (full set) ===

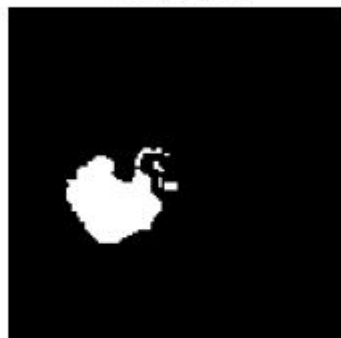
Mean Dice: 0.6706 | Mean IoU: 0.5762

Precision: 0.8896 | Recall: 0.7737 | F1: 0.8276

Original



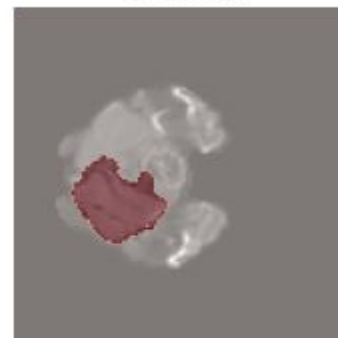
GT Mask



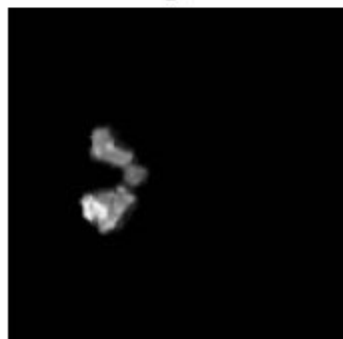
Predicted



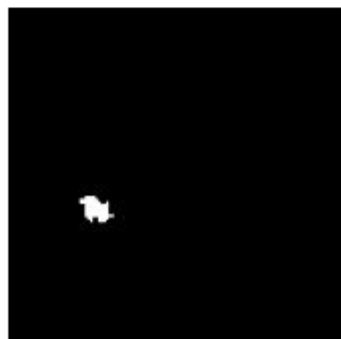
Overlay



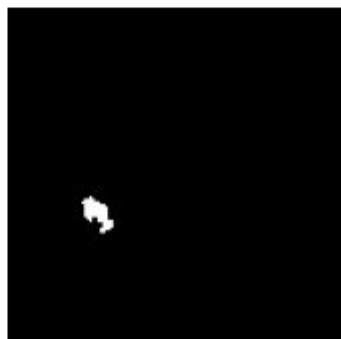
Original



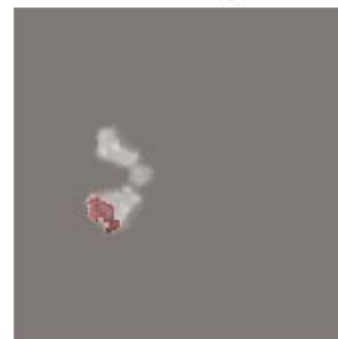
GT Mask



Predicted



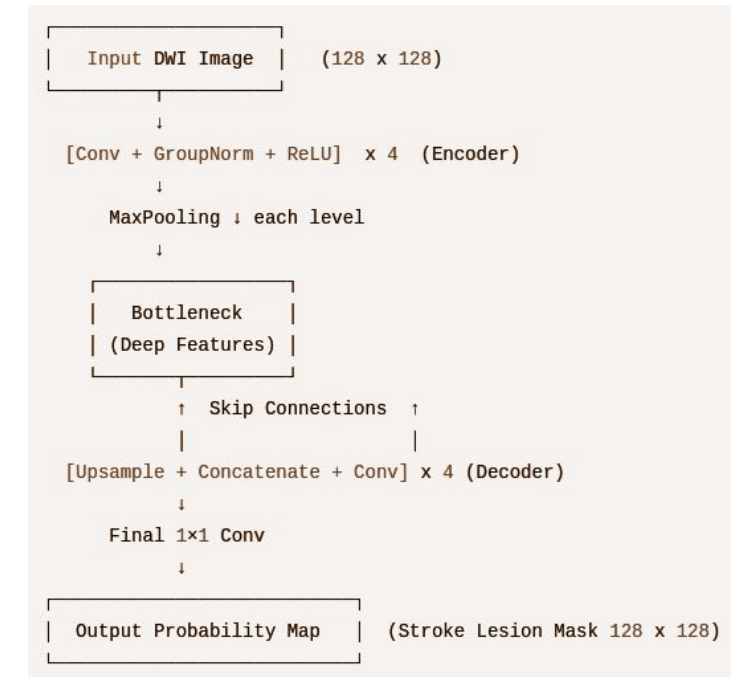
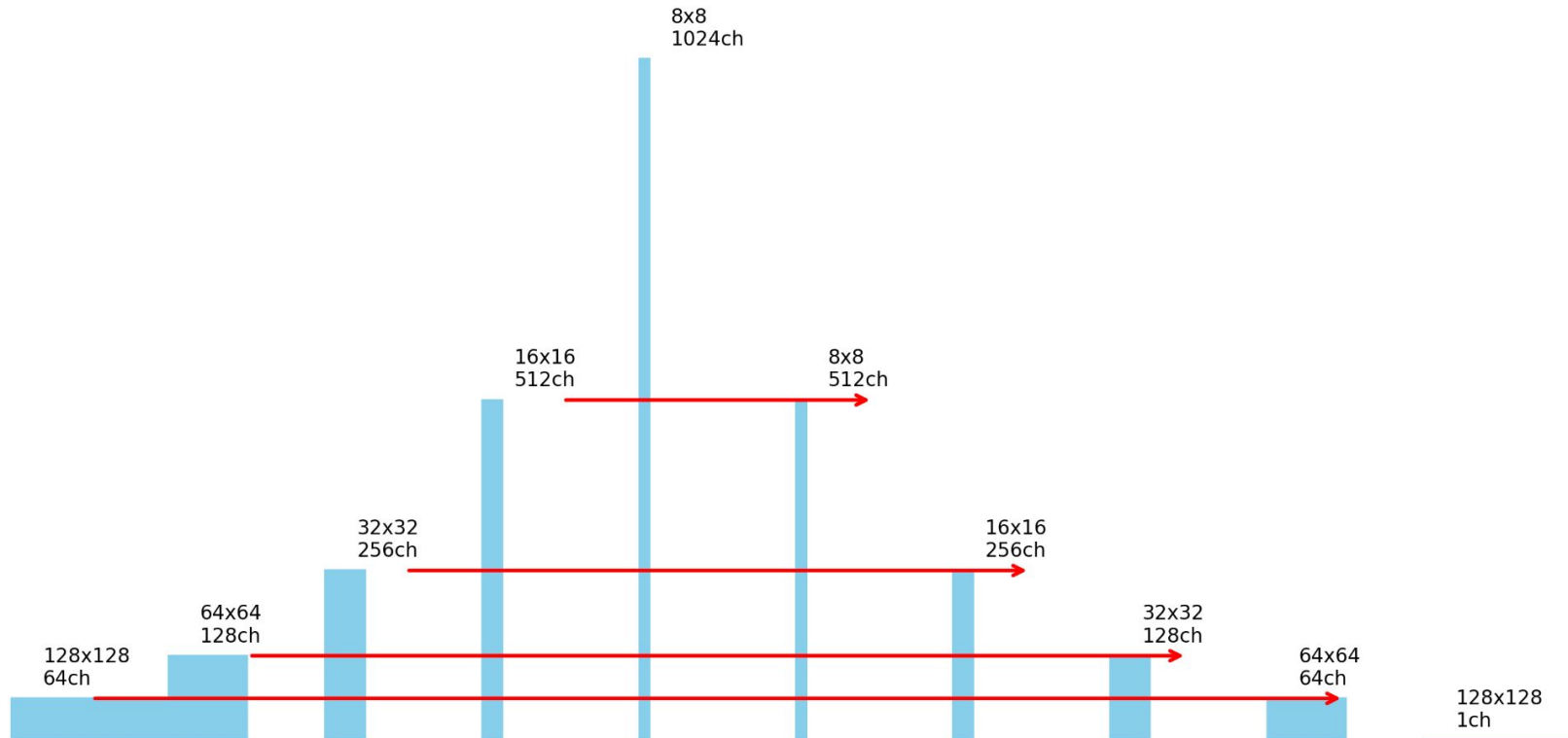
Overlay



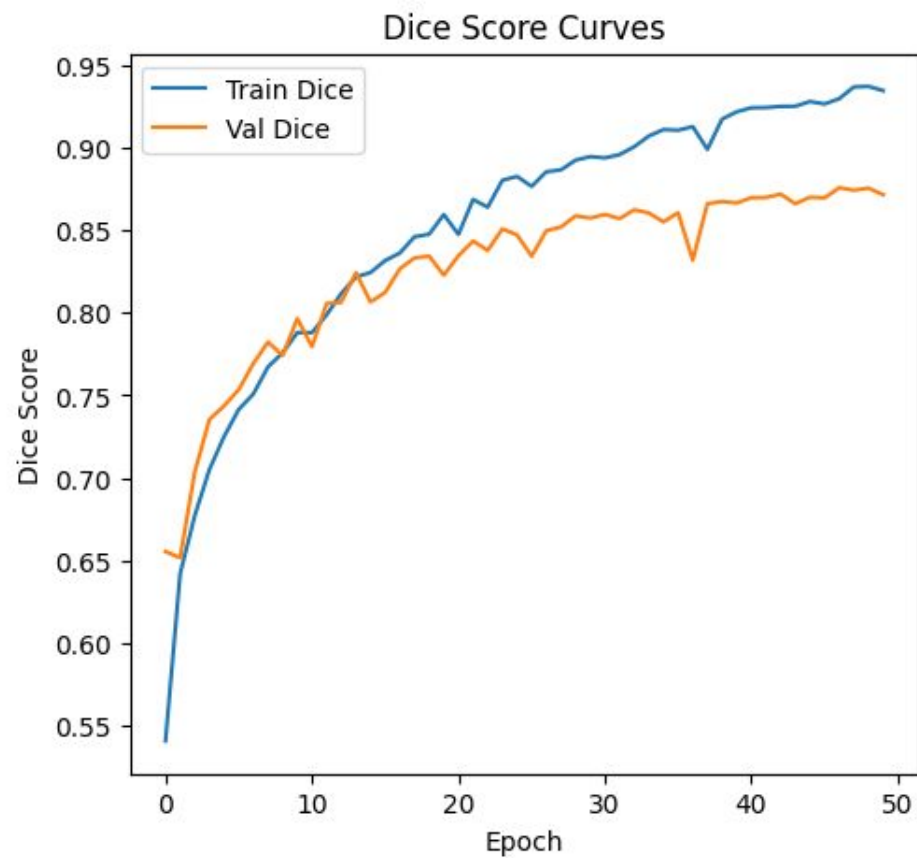
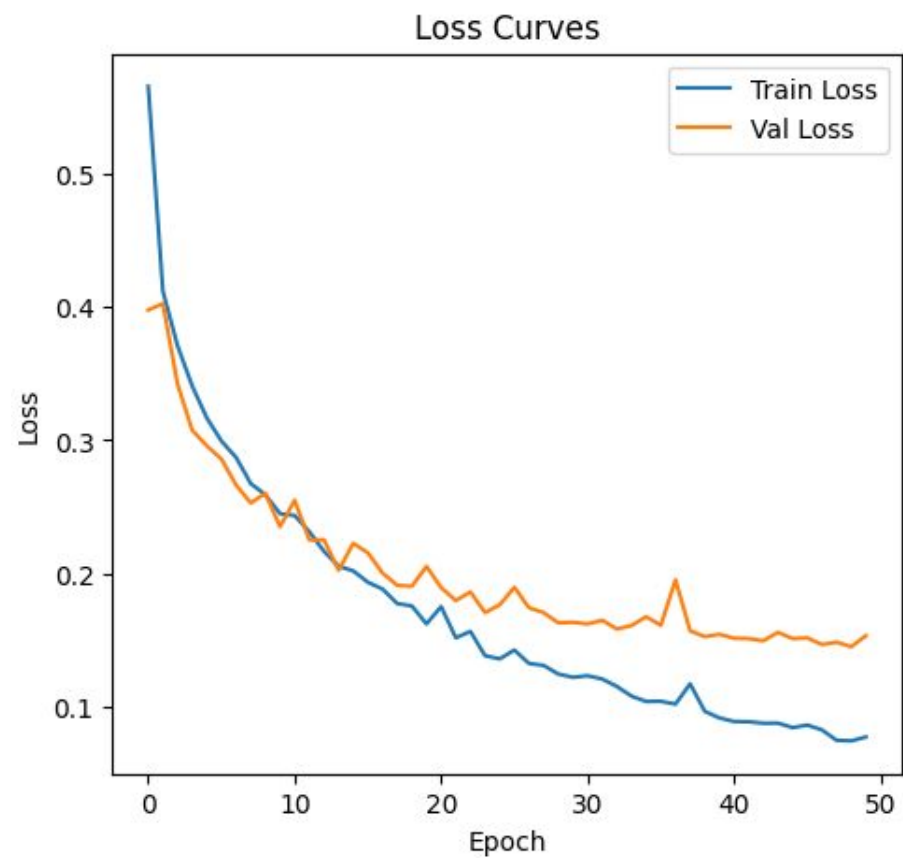
Best Model



UNet Feature Map Diagram with Skip Connections



Modified Unet Model



Predictions



Test Metrics:

Average Dice: 0.89

Average IoU: 0.81

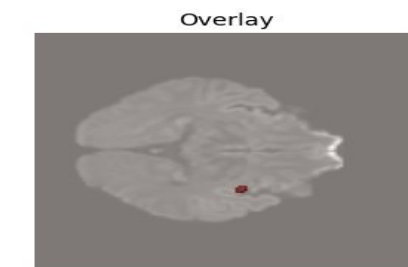
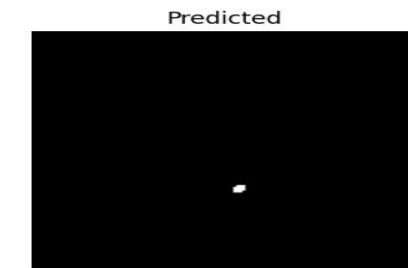
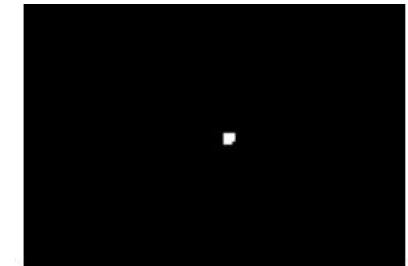
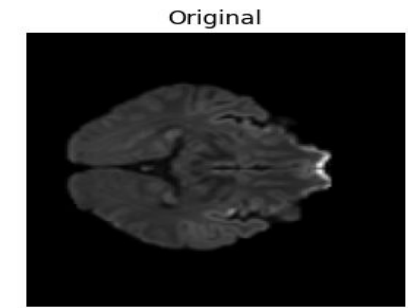
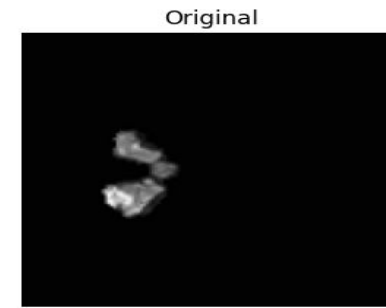
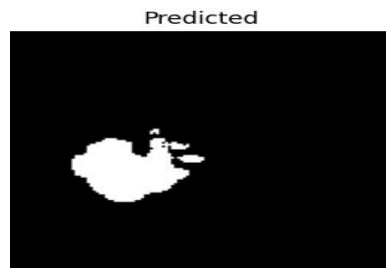
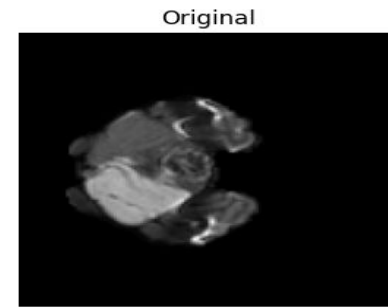
Precision: 0.9410,

Recall: 0.94,

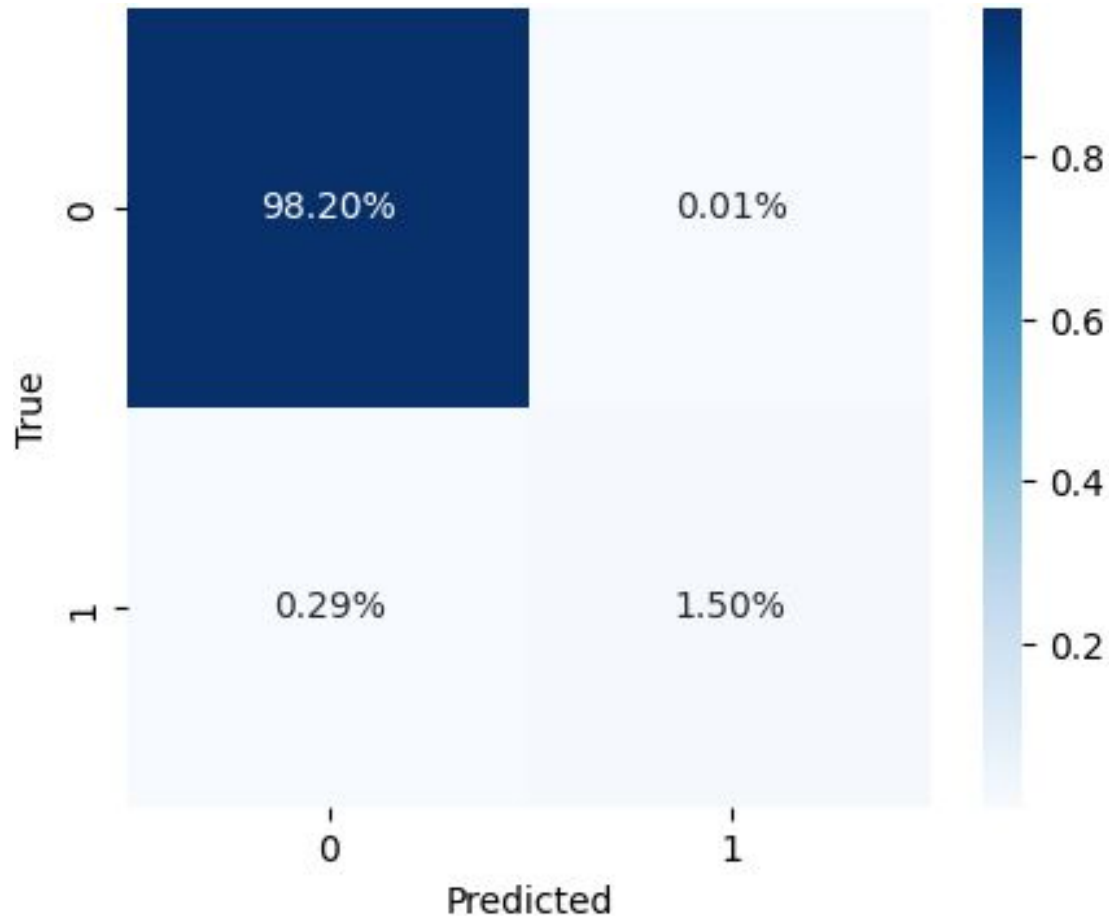
F1-score: 0.94

ISLES 2022 benchmark:

DWI alone Dice 83.9%,



Confusion Matrix



Due to the pixel difference between background and ground truth, high accuracy score does not give meaningful lesion prediction.

Limitations



- Uses **only DWI**, missing complementary info from ADC, FLAIR, or T2.
- May **miss very small or subtle infarcts**.
- Trained only on ischemic stroke → **limited generalizability** to other brain pathologies.

Future Work



- **Integrate multi-modal MRI** (ADC, FLAIR, T2) for improved lesion characterization.
- **Enhance small infarct detection** through higher resolution or specialized loss functions.
- **Expand dataset** to include other brain pathologies (hemorrhage, tumors) for better generalizability.



Key Takeaways

- **DWI-based deep learning** can segment ischemic stroke lesions with good accuracy.
- **Multi-modal integration** (ADC, FLAIR, T2) could further improve performance.
- **Small infarcts and generalizability** remain challenges for clinical application.



Acknowledgements



- I sincerely thank **Moffitt Cancer Center** for providing this invaluable internship opportunity.
- My gratitude goes to **Dr. Eduardo G. Moros** for his support.
- Special thanks to my mentor, **Dr. Tess Armstrong**, for her kindness, respect, encouragement, and invaluable mentorship throughout this work.



Thank You



Any Questions?

Actively seeking a postdoctoral position in health and disease research

pawanthapaliya09@gmail.com

Variable Definitions



H_{in}, W_{in} : Height and width of the input feature map.

C_{in} : Number of input channels.

K : Kernel size (assume square, $K \times K$).

C_{out} : Number of output channels (number of filters).

P : Padding.

S : Stride.

H_{out}, W_{out} : Output height and width.

$X \in \mathbb{R}^{H_{in} \times W_{in} \times C_{in}}$: Input tensor.

$W \in \mathbb{R}^{K \times K \times C_{in} \times C_{out}}$: Convolution weights.

$b \in \mathbb{R}^{C_{out}}$: Convolution bias.

$Y \in \mathbb{R}^{H_{out} \times W_{out} \times C_{out}}$: Output tensor.

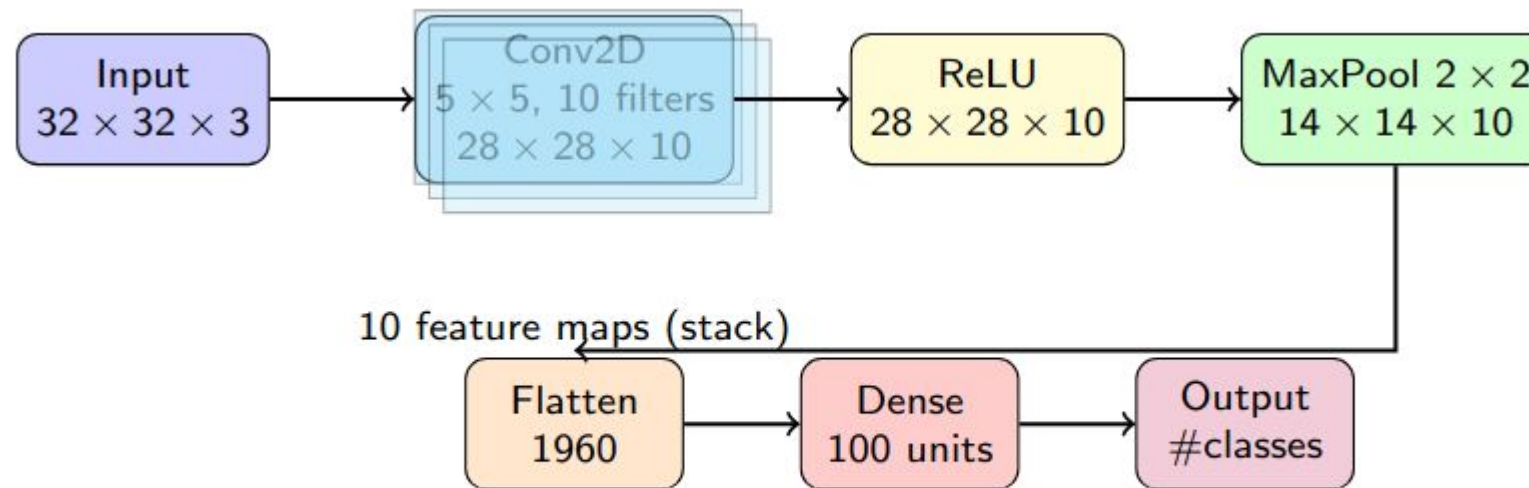
$f(\cdot)$: Activation function (e.g., ReLU $f(x) = \max(0, x)$).

$X_{\text{flat}} \in \mathbb{R}^{N_{in}}$: Flattened input.

N_{out} : Number of neurons in fully connected layer.

$W_{fc} \in \mathbb{R}^{N_{out} \times N_{in}}, b_{fc} \in \mathbb{R}^{N_{out}}$: Fully connected weights and biases.

CNN Architecture with 10 Feature Maps





Convolution Output Size Formula

Output size:

$$H_{out} = \left\lfloor \frac{H_{in} - K + 2P}{S} \right\rfloor + 1, \quad W_{out} = \left\lfloor \frac{W_{in} - K + 2P}{S} \right\rfloor + 1$$

Convolution operation:

$$Y_{i,j,k} = \sum_{c=0}^{C_{in}-1} \sum_{m=0}^{K-1} \sum_{n=0}^{K-1} X_{iS+m-P, jS+n-P, c} \cdot W_{m,n,c,k} + b_k$$

Number of parameters:

$$\text{Params} = K \times K \times C_{in} \times C_{out} + C_{out}$$

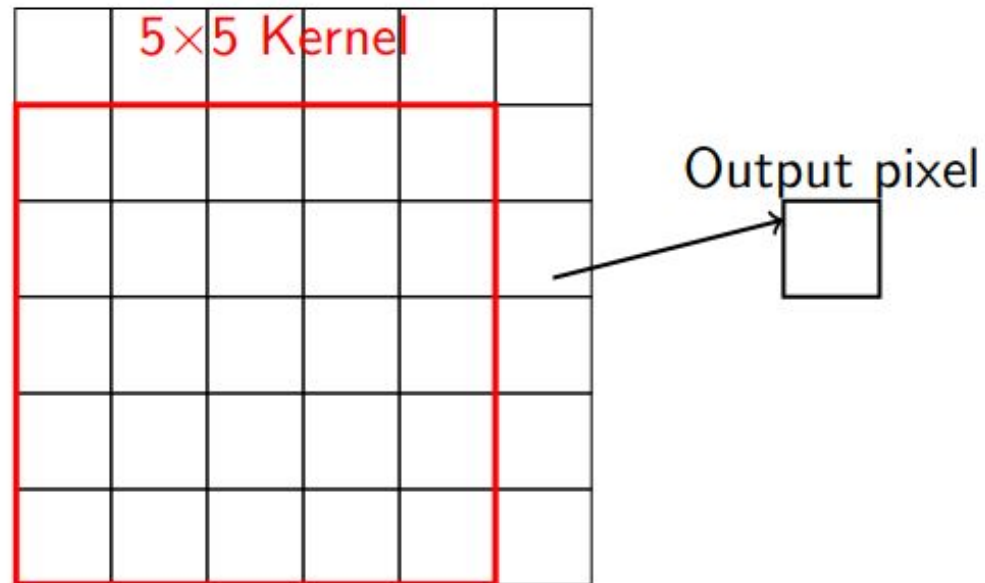


Convolution Example

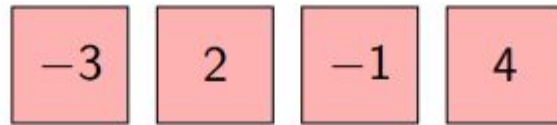
Input: 6×6 , Kernel: 5×5 , $S = 1, P = 0$

Output size: $\frac{6 - 5 + 0}{1} + 1 = 2 \Rightarrow 2 \times 2$

Input 6×6



Relu Activation Functions



Input Feature Map



Output after ReLU: $f(x) = \max(0, x)$



Max-Pooling Formula and Example

Output size:

$$H_{out} = \left\lfloor \frac{H_{in} - K}{S} \right\rfloor + 1, \quad W_{out} = \left\lfloor \frac{W_{in} - K}{S} \right\rfloor + 1$$

Example: Input $28 \times 28 \times 10$, MaxPool 2×2 , stride 2:

$$H_{out} = W_{out} = \frac{28 - 2}{2} + 1 = 14$$

Small numeric example:

$$\max \begin{bmatrix} 4 & 9 \\ 1 & 7 \end{bmatrix} = 9$$



Flatten and Dense Parameters

Flatten: Converts $14 \times 14 \times 10$ to $X_{\text{flat}} \in \mathbb{R}^{1960}$

Fully connected layer:

$$Y = W_{fc} \cdot X_{\text{flat}} + b_{fc}, \quad \text{Params} = N_{in} \cdot N_{out} + N_{out}$$

Example: Flatten 1960, Dense 100 neurons:

$$\text{Params} = 1960 \cdot 100 + 100 = 196100$$



Derivative wrt prediction:

$$\frac{\partial L_i}{\partial \hat{y}_i} = -\frac{y_i}{\hat{y}_i} + \frac{1 - y_i}{1 - \hat{y}_i}$$

Sigmoid derivative:

$$\frac{\partial \hat{y}_i}{\partial z_i} = \hat{y}_i(1 - \hat{y}_i)$$

Chain rule (main gradient):

$$\frac{\partial L_i}{\partial z_i} = \hat{y}_i - y_i$$

Parameter gradients:

$$\frac{\partial L_i}{\partial w} = (\hat{y}_i - y_i)x_i$$

$$\frac{\partial L_i}{\partial b} = \hat{y}_i - y_i$$



Gradient Descent Updates

Update rules:

$$w \leftarrow w - \eta \frac{\partial L}{\partial w}$$

$$b \leftarrow b - \eta \frac{\partial L}{\partial b}$$

Training steps:

1. Forward pass: compute z_i and $\hat{y}_i$
2. Compute BCE loss
3. Backprop: compute $\hat{y}_i - y_i$
4. Compute gradients for w, b
5. Update parameters